

Author's Greeting

Thank you for reading this Technical Note accompanying the KFW Original Squat Definition Document.

At first glance, the squat appears to be a simple movement. However, it is also an exercise that readily exposes the structural vulnerabilities inherent in the human body. In particular, during lower-body movement, the lumbar spine, pelvis, hip joints, and knee joints are placed in positional relationships that are prone to instability.

From an early age, I intuitively recognized that avoiding lower back pain during standing and sitting requires deliberately offsetting the timing of hip and knee flexion. Yet, no matter how many books on squatting I searched for, I could not find any that addressed this point directly.

I continued searching for some time, but eventually stopped looking altogether and decided instead to construct the squat myself.

That decision led to the creation of the KFW Original Squat.

The KFW Original Squat was developed not through reliance on intuition or tradition, but through a commitment to structural clarity in addressing these vulnerabilities.

In this Technical Note, I present the concept of structural fixation as the core principle governing the lower-body structure of the squat.

This principle emerged from careful observation, repeated verification, and a consistent insistence on preserving the integrity of the human body under load.

It is my hope that this note will deepen your understanding of the KFW Original Squat and provide a clear structural foundation for both practice and instruction.

Preface

The lower half of the squat has long been treated as a region governed by intuition, experience, or tradition rather than structural clarity. Yet this portion of the movement exposes the body to its greatest mechanical vulnerability. Lumbar flexion, posterior pelvic tilt, anterior glide of the hip joint, and rotational stress at the knee do not arise from mistakes or poor technique alone. They emerge from the inherent limitations of human structure.

The KFW Original Squat was developed to address these limitations through design rather than compensation. Its purpose is not to teach a “style” of squatting, but to articulate a structural solution—one that preserves the integrity of the spine and hips under load, and one that can be reproduced with precision.

This technical supplement focuses on a single, indispensable principle within that design: structural fixation in the lower half. It clarifies why the lower half must be approached differently from the upper half, why the hip must complete its hinge before the knee assumes responsibility for depth, and why the descent and ascent must mirror each other to maintain structural integrity.

The pages that follow do not introduce new ideas beyond the Definition Document. Instead, they illuminate the structural logic already embedded within the KFW Original Squat, providing a deeper understanding of the movement's most delicate phase. My intention is that this supplement serves as a bridge—linking the conceptual clarity of the Definition Document with the practical demands of real-world execution.

This Technical Note addresses the lower half of the squat as a single, continuous structural event. Within this framework, the descent, bottom position, and ascent are not treated as separate phases, but as consecutive expressions of one fixed structural condition. The Full Hip Hinge, established during the descent at the half-squat position, is maintained without alteration through the bottom and into the ascent until the structure exits the lower half. This document therefore examines not isolated movements, but the conditions under which structural continuity is preserved across the entire lower-half range of motion.

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Chapter 1

The Structural Vulnerability of the Lower Half

The lower half of the squat represents the most structurally vulnerable region of the entire movement. As the body descends past the midpoint, the lumbar spine, pelvis, hip joints, and knees enter a configuration in which their inherent mechanical limitations become increasingly exposed. These vulnerabilities are not the result of poor technique, insufficient strength, or lack of mobility. They arise from the fundamental architecture of the human body.

The lumbar spine, when placed under load, has a natural tendency toward flexion. The pelvis, influenced by the orientation of the femur and the tension of surrounding musculature, drifts toward posterior tilt. The hip joint, constrained by the shape of the acetabulum and the direction of force, is prone to anterior glide. The knee, positioned between two large rotational structures, becomes susceptible to torsional stress.

These phenomena occur not because the lifter “fails,” but because the structure itself demands clarity. Without a deliberate strategy to stabilize these regions before entering the lower half, the body

will default to its path of least resistance. This path, while mechanically efficient in the short term, compromises the integrity of the spine and hips under load.

For this reason, the lower half cannot be approached as a simple continuation of the upper half. It requires a distinct structural strategy—one that prepares the body before depth is created, and one that prevents the collapse of the spine–pelvis–hip complex. The KFW Original Squat addresses this requirement through the principle of structural fixation, which ensures that the body enters the lower half with a stable, predetermined configuration.

Understanding the inherent vulnerability of this region is the first step toward mastering the structural logic of the squat. The chapters that follow will clarify why the hip must complete its hinge before the knee assumes responsibility for depth, and how this sequence protects the body from the mechanical risks that emerge in the lower half.

Chapter 2

Why the Hip Must Fold Before the Knee

A structurally sound squat requires that the hip initiates the descent before the knee assumes responsibility for depth. This sequence is not a matter of style or preference; it is a structural necessity. The hip joint, with its capacity for large rotational movement and its role as the primary hinge of the lower body, must establish the foundation upon which the rest of the descent is built.

If the knee initiates the descent prematurely, the pelvis is forced into a compromised orientation. The lumbar spine loses its neutral alignment, the pelvis drifts toward posterior tilt, and the hip joint begins to glide anteriorly. These changes occur subtly at first, but under load they accumulate into a pattern that destabilizes the entire spine–pelvis–hip complex.

Conversely, when the hip folds first—within a safe and controlled range—it creates the structural conditions necessary for a stable descent. This initial hinge allows the pelvis to maintain anterior tilt, the lumbar spine to remain neutral, and the hip joint to stay

centered within the acetabulum. The knee then bends in coordination with the hip, not in competition with it.

However, the hip cannot be folded deeply from the outset. A rapid or excessive hinge leads to an exaggerated forward lean, commonly known as the “good morning squat,” which shifts the load disproportionately onto the lumbar spine. The solution is a progressive hinge: the hip folds first, but only to the degree that preserves balance, spinal integrity, and the ability to continue descending without structural collapse.

This progressive hinge culminates at the half-squat position, where the hip and knee reach approximately ninety degrees. At this point, the hip has completed its hinge, and the responsibility for creating further depth transitions to the knee. This transition is not arbitrary—it is dictated by the limits of human structure.

Understanding why the hip must fold before the knee clarifies the logic of the entire descent. It ensures that the body enters the lower half with a stable configuration, prepared for the structural demands that follow.

Chapter 3

Half Squat as the Completion of Full Hip Hinge

The half-squat position marks the completion of the Full Hip Hinge, a structural milestone that determines the integrity of the remainder of the descent. At approximately ninety degrees of hip and knee flexion, the hip joint reaches the limit of its safe and functional hinge. Beyond this point, the hip cannot fold further without compromising the alignment of the pelvis and lumbar spine.

This completion of the hinge is not a matter of sensation or subjective depth. It is a structural event. When the hip reaches its hinge limit, the pelvis stabilizes in anterior tilt, the lumbar spine maintains its neutral alignment, and the femoral head remains centered within the acetabulum. These conditions form the foundation upon which the lower half of the squat must be built.

If the lifter attempts to continue folding at the hip beyond this point, the structure begins to deteriorate. The pelvis drifts toward posterior tilt, the lumbar spine begins to flex, and the hip joint glides forward. These changes are subtle at first, but under load

they accumulate rapidly, leading to a collapse of the spine–pelvis–hip complex.

For this reason, the half-squat position serves as the structural threshold of the movement. It is the final moment in which the hip contributes to depth. Once the hinge is complete, the responsibility for creating further descent transitions entirely to the knee. This transition is not optional; it is dictated by the architecture of the human body.

Recognizing the half squat as the completion of the Full Hip Hinge allows the lifter to enter the lower half with clarity and stability. It ensures that the descent proceeds not through improvisation, but through a predetermined structural sequence. The chapters that follow will explain how this sequence gives rise to the paradox of the lower half and why structural fixation becomes essential once the hinge is complete.

Chapter 4

The Structural Paradox of the Lower Half

The lower half of the squat presents a unique structural paradox—one that cannot be resolved through strength, flexibility, or technique alone. It emerges from the fundamental architecture of the hip joint and the spine–pelvis complex, and it defines the precise moment at which the descent must change its governing mechanism.

The paradox is this:

The hip can no longer fold,
yet the lifter must continue applying the intention to fold.

Once the hip reaches the limit of its hinge at the half-squat position, further flexion is structurally impossible without compromising alignment. Attempting to fold the hip beyond this point forces the pelvis toward posterior tilt, draws the lumbar spine into flexion, and causes the femoral head to glide anteriorly within the acetabulum. These changes destabilize the entire structure and place the spine at risk under load.

Yet paradoxically, the intention to fold must not disappear.

If the lifter abandons the hinge intention at the moment the hinge becomes mechanically impossible, the pelvis loses its anterior orientation, the lumbar spine collapses, and the hip joint shifts forward. The structure deteriorates not because the lifter moved incorrectly, but because the intention that maintains the configuration was released.

This is the essence of the structural paradox:

The hip must remain fixed in angle, but the internal directive to hinge must continue.

This sustained intention stabilizes the pelvis, preserves lumbar neutrality, and prevents anterior glide of the hip. It transforms the lower half from a region of vulnerability into a region of controlled structural descent.

Once the hinge is complete, the knee becomes the sole contributor to depth. The hip angle remains fixed, yet the hinge intention remains active. This duality—fixed structure with sustained intention—is what allows the lifter to enter the lower half without compromising the spine–pelvis–hip complex.

Understanding this paradox is essential for mastering the KFW Original Squat. It clarifies why the lower half cannot be approached through intuition or sensation alone, and why structural fixation becomes the defining principle of this phase of the movement.

Chapter 5

Structural Fixation in the Lower Half

Once the hip has completed its hinge at the half-squat position, the descent enters a fundamentally different structural phase. The hip angle can no longer change without compromising the alignment of the pelvis and lumbar spine. For this reason, the lower half of the squat must be governed by a single, non-negotiable principle:

The hip angle remains fixed,
and depth is created exclusively through the knee.

This fixed-hip configuration is not a stylistic choice. It is a structural requirement dictated by the limits of human anatomy. When the hip angle is held constant, the pelvis maintains anterior tilt, the lumbar spine preserves its neutral alignment, and the femoral head remains centered within the acetabulum. These conditions prevent the collapse that typically occurs when lifters attempt to “sink” deeper by continuing to fold at the hip.

Structural fixation transforms the lower half from a region of vulnerability into a region of controlled descent. The knee

becomes the sole joint responsible for creating additional depth, while the hip maintains the configuration established at the completion of the hinge. This division of responsibility—hip for structure, knee for depth—is the defining characteristic of the KFW Original Squat.

If the hip angle changes during the lower half, even slightly, the structure deteriorates. The pelvis drifts toward posterior tilt, the lumbar spine begins to flex, and the hip joint glides forward. These changes occur rapidly under load and cannot be corrected mid-descent. Structural fixation prevents this cascade by eliminating unnecessary movement at the hip and preserving the integrity of the spine–pelvis–hip complex.

The lower half is therefore not a continuation of the upper half. It is a distinct structural environment that requires a fixed configuration and a clear division of mechanical roles. The hip stabilizes; the knee moves. The intention to hinge remains active, but the angle does not change. This is the essence of structural fixation.

Understanding and applying this principle is essential for executing the KFW Original Squat safely and effectively. It ensures that the lifter descends with clarity, maintains structural integrity under load, and prepares the body for an ascent that mirrors the descent.

Chapter 6

Ascent Mirrors Descent

The ascent is not the construction of a new structure, nor the initiation of a different technical phase.

It is the controlled release of a structure that was already fixed during the descent.

Accordingly, no new structural rules are introduced during the ascent.

The Full Hip Hinge established at the half-squat position during the descent remains the sole structural reference through the bottom and into the ascent.

As the ascent begins from the full-bottom position, knee extension occurs while the hip angle remains unchanged.

The intent during ascent is not to open the hinge, but to prevent its loss.

The structurally most critical interval of the ascent lies between the full-bottom position and the half-squat.

Within this range, particularly near the parallel position often recognized as the sticking point,

the integrity of the previously established structure is most clearly revealed.

Any apparent deceleration or difficulty at this point does not indicate a new problem, but rather exposes whether the fixed structure has been preserved.

If the hinge angle collapses during this interval, structural continuity is broken.

If the hinge angle is maintained through the sticking region and back to the half-squat, the structure established during the descent is confirmed to have remained intact.

The half-squat position serves as the sole structural boundary for the lower half during ascent, just as it does during the descent.

Once this position is reached, the role of the fixed lower-half structure is complete, and any subsequent movement lies outside the scope of this Technical Note.

Chapter 7

Summary: The KFW Structural Principle of the Lower Half

The structural logic of the lower half of the squat can be distilled into a single, essential principle:

Complete the Full Hip Hinge before entering the lower half,
fix the hip angle at that point,
and create depth exclusively through the knee while maintaining
hinge intention.

This principle resolves the inherent vulnerabilities of the lower half by establishing a stable configuration before the body enters its most mechanically delicate region. Once the hip has reached the limit of its hinge at the half-squat position, the pelvis stabilizes in anterior tilt, the lumbar spine maintains its neutral alignment, and the hip joint remains centered within the acetabulum. These conditions form the structural foundation for safe and controlled descent.

From this point forward, the hip angle must remain fixed. The knee becomes the sole contributor to depth, while the intention to hinge continues internally to preserve the configuration. This

duality—fixed structure with sustained intention—prevents the collapse of the spine–pelvis–hip complex and transforms the lower half from a region of vulnerability into a region of structural clarity.

The ascent mirrors the descent. The knee initiates the rise, the hip angle remains fixed until the half-squat position is regained, and the hinge intention remains active throughout. This symmetry ensures that the squat is executed as a single, continuous structural event rather than a sequence of disconnected phases.

The KFW Original Squat is defined not by style, sensation, or tradition, but by structure. The principle outlined in this supplement clarifies the logic of the lower half and provides a reproducible method for preserving the integrity of the human body under load. It is a principle that elevates the squat from a movement pattern to a structural discipline.

Taken together, the descent, bottom position, and ascent of the lower half constitute a single closed structural event,

defined by early fixation of the Full Hip Hinge and its uninterrupted preservation until the half-squat boundary is reached again.

Chapter 8

Technical Considerations of the KFW Original Squat

Understanding the structure of the KFW Original Squat hinges on two core principles: the priority of hip flexion and the completion of the Full Hip Hinge at the half-squat position.

The purpose of this chapter is to organize, from a technical perspective, the conditions under which this judgment was examined, the process through which it was refined, and the reasoning that led to its fixation as the defining structure of the lower half.

In the KFW Original Squat, the descent is not regarded as a simple flexion–extension movement.

In particular, the transition from what is often referred to as a “half good morning” to the half-squat is not treated as an intermediate phase of motion, but as a clearly defined setup period for establishing structural stability.

The priority of hip flexion during this interval is an indispensable condition for preventing knee-dominant compensatory patterns and for preferentially recruiting the posterior chain.

This hip-initiated movement strategy is widely shared within the fields of rehabilitation and sports medicine and is consistent with standard approaches aimed at reducing shear forces at the knee joint and patellofemoral stress.

What the KFW method clarifies further is the definition of the half-squat position as the sole structural boundary of the lower half.

Defining the half-squat as the point at which the individual completes the Full Hip Hinge—representing the maximum depth at which spinal alignment can be maintained—is a deliberate and intentional demarcation.

Once established, this hinge angle is not altered during subsequent movement.

Throughout the mid-squat section, from the half-squat to the full-bottom position and back again, ascent and descent are performed solely through knee flexion and extension while maintaining this hinge angle.

This concept of hinge-held is treated not as a technical cue, but as a fixed structural condition.

In practical terms, this squat completes the Full Hip Hinge at the half-squat position during the descent.

From there, the lifter continues descending to the bottom position and initiates the ascent while preserving the same hinge angle.

During the ascent from the full bottom, this hinge angle remains unchanged as the movement passes through the region commonly recognized as the sticking point near parallel, and continues until the half-squat position is reached again.

Throughout this entire sequence, a single, consistent structure is maintained.

The hinge is not something that is constructed gradually along the way.

It is already completed at the half-squat during the descent and is subsequently tested across every phase of movement.

This continuous verification of the same structure constitutes the most critical aspect of the KFW Original Squat.

Because the hinge angle is fixed early and preserved from the full-bottom position through the ascent—including the sticking region—this squat can be said to inherently include a mechanism for preventing butt wink.

This is not a methodology aimed at correcting a specific phenomenon, but rather the outcome of a design decision that establishes a structure in which such phenomena cannot occur in the first place.

The considerations presented in this chapter clarify the technical basis for why the lower half of the KFW Original Squat is treated as a structure that is not subject to adjustment, interpretation, or individualized modification.

What is offered here is neither a set of practical instructions nor a restatement of definitions.

Instead, this chapter serves as a technical examination that makes explicit the prerequisite conditions, evaluative process, and reasoning that demonstrate this structure was not chosen arbitrarily, but as a necessary conclusion.

Afterword

The principles presented in this supplement were not discovered through theory alone. They emerged from years of personal exploration—through discomfort, trial, error, and the persistent search for structural clarity. The lower half of the squat, often treated as a matter of intuition or experience, revealed itself to be a region governed by precise mechanical laws. Understanding these laws required patience, observation, and a willingness to question long-held assumptions.

The KFW Original Squat was not created to add another variation to the landscape of training methods. Its purpose is to articulate a structural framework that protects the human body under load and clarifies the logic of movement. The concept of structural fixation in the lower half is one of the most important elements of that framework. It transforms the squat from a task of strength into a discipline of structure.

My hope is that this document serves not only as a technical explanation, but also as an invitation—to observe the body more closely, to question what seems familiar, and to approach movement with a deeper sense of responsibility. Structure is not

merely a constraint; it is a form of freedom. When the body is aligned with its own architecture, movement becomes both safer and more powerful.

Thank you for taking the time to engage with this work.

May it support your practice, your teaching, and your ongoing exploration of human movement.

Author's Note

The content presented in this note did not arise from a short period of consideration.

Rather, it is the result of many years of personal trial and reflection.

These ideas were neither inherited from an existing method nor discovered all at once.

Beginning with bodily intuitions I have held since childhood, they gradually became clearer through sustained observation of human movement, questioning of established assumptions, and a continued search for structures that remain internally consistent.

The KFW Original Squat is part of a broader effort to articulate a coherent cultural and technical system that respects the structure of the human body and enables reproducible practice.

My intention has never been to propose a new training style, but to clarify the fundamental structures that govern safe and efficient movement.

The primary objectives of the KFW Original Squat can be distilled into two points.

- To maintain the natural curvature of the spine throughout all movement,
while preserving mobility of the upper body.
- To complete the Full Hip Hinge during the descent of the squat and, by preserving it, to establish a robust and reliable movement linkage
that protects the lower back.

To practice the KFW Original Squat is not merely to understand these principles,
but to repeatedly learn, verify, and embody them through the body.

This note was included because the structure of the lower half of the squat
required an explanation in its own right.

The structural paradox, the completion of the Full Hip Hinge,

and the necessity of fixation are not incidental details.
They are essential elements for correctly understanding a
movement
that has been misunderstood across generations.
Recording these elements is part of my responsibility
as a steward entrusted with this framework.

I hope that this note will further deepen understanding of the
KFW system
and encourage engagement with movement grounded in both
precision and curiosity.
Structure is not merely a technical concept;
it is a perspective for understanding the body,
and ultimately, a means of understanding oneself.

Satoru Hara